

# A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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## Introduction

- Multi-compartment models represent the spatial morphology of a neuron as connected compartments, reproducing neuronal activity in fine detail.
- Each compartment is described by differential equations carrying **gate variables** (ion-channel open probabilities), as in the Hodgkin–Huxley (HH) model.

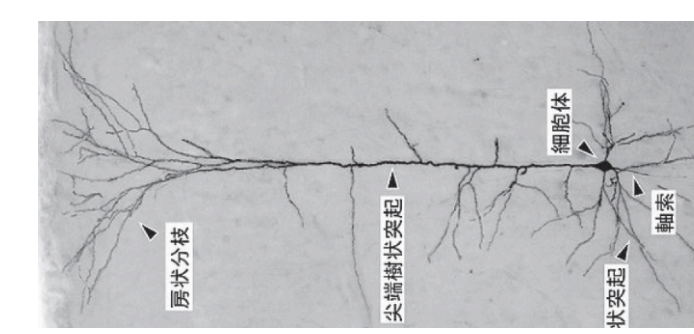
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{Na}m^3h(V - E_{Na}) - \bar{g}_Kn^4(V - E_K) + I_{\text{ext}}$$

HH membrane-potential equation ( $m, h, n$ : gate variables)

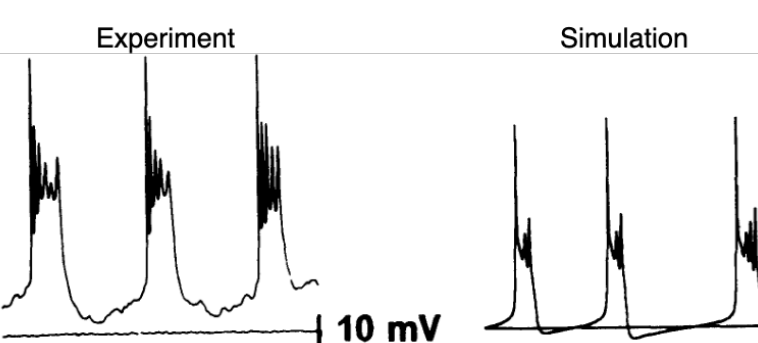
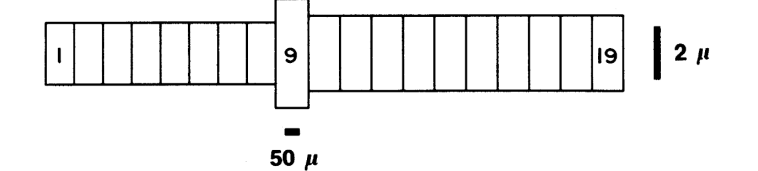
→ When many multi-compartment models are simulated in parallel (e.g. brain-scale simulation), the gate variables cause a **memory bottleneck**.

### Goal

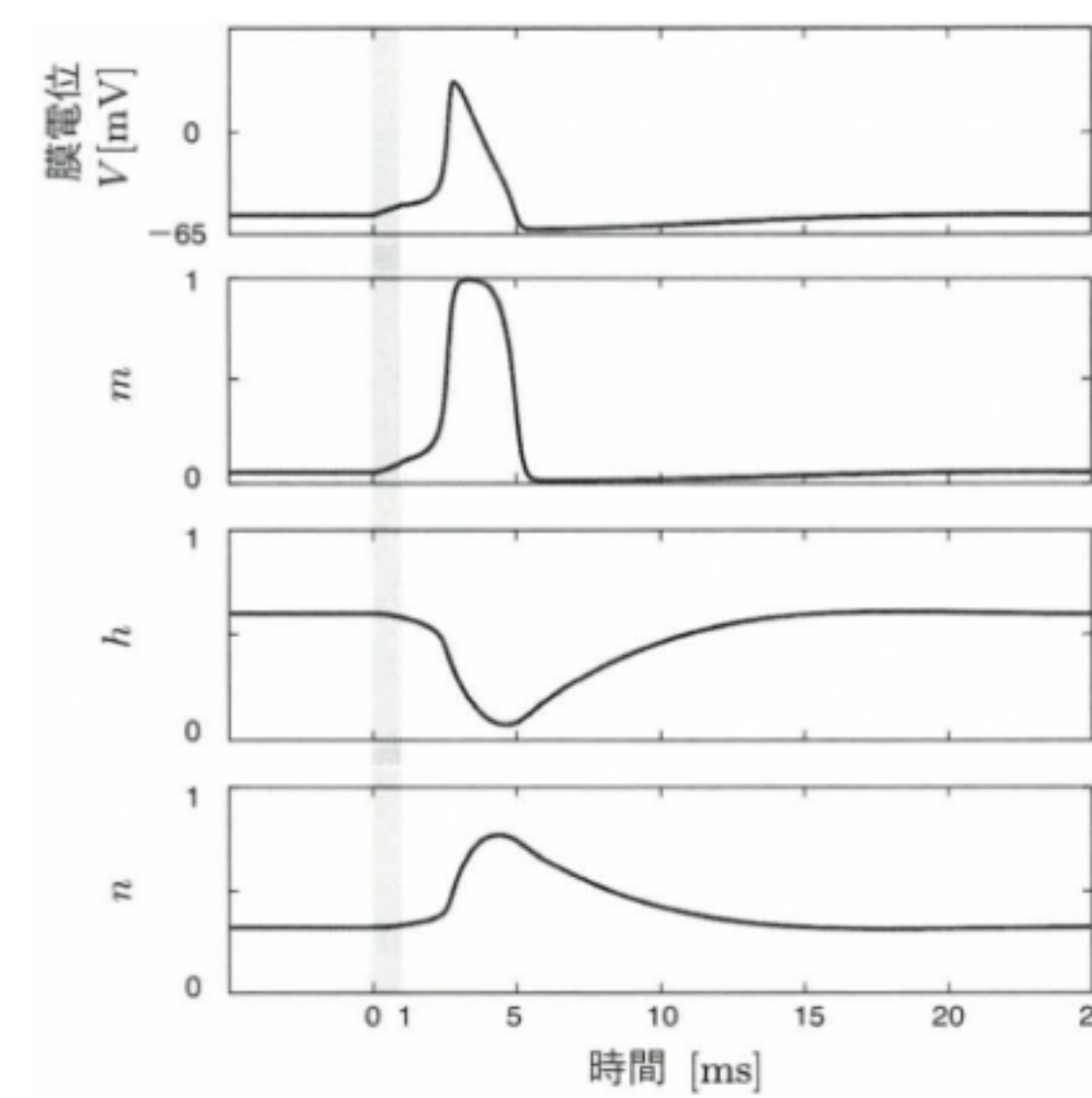
Build a surrogate model that reproduces the membrane-potential response of a multi-compartment neuron with **fewer state variables**.



Pyramidal neuron [1]



Spatial representation and simulation of a multi-compartment model [2]



HH membrane-potential response [3]

## Methods

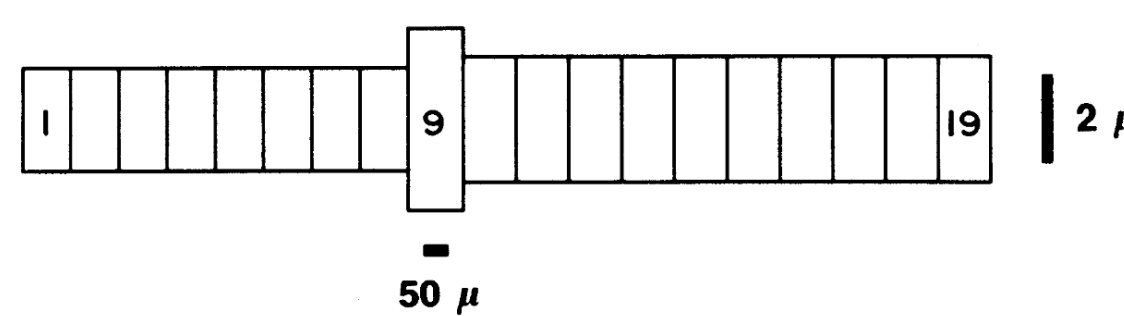
### Two-axis surrogate framework

- Simulate the target; collect  $V + \text{gate time series}$ .
- Compress** gates into a low-dim latent  $z \in \mathbb{R}^n$  via a **preprocessor** — PCA (linear) or an **autoencoder** (nonlinear).
- Identify** the latent dynamics. The **ansatz** sets the form: **SINDy** picks from a generic library; **hybrid** keeps the ionic physics, fitting **only** the latent.

$$\begin{cases} \frac{dV}{dt} = \underbrace{I_{\text{ion}}^{\text{phys}}(V, \mathbf{z})}_{\text{known physics}} + I_{\text{ext}} \\ \frac{dz}{dt} = \underbrace{\Theta(\mathbf{z}, V)\boldsymbol{\xi}}_{\text{SINDy-identified}} \\ \mathbf{z} = \text{AE}_{\text{enc}}(\text{gates}), \quad \text{gates} = \text{AE}_{\text{dec}}(\mathbf{z}) \end{cases}$$

**hybrid** fixes last year's **over-large** (41-term) SINDy library.

**Target: Traub CA3 pyramidal cell** — a 19-compartment realistic neuron (vs. last year's toy 3-comp).



**Replace the soma with the surrogate:** soma gates →  $\mathbf{z}$ ; compartment coupling stays physical.

- preprocessor  $\in \{\text{PCA}, \text{AE}\}$
- ansatz  $\in \{\text{SINDy}, \text{hybrid}\}$
- latent dim  $n = 2$  (HH) / 3 (Traub)

### SINDy

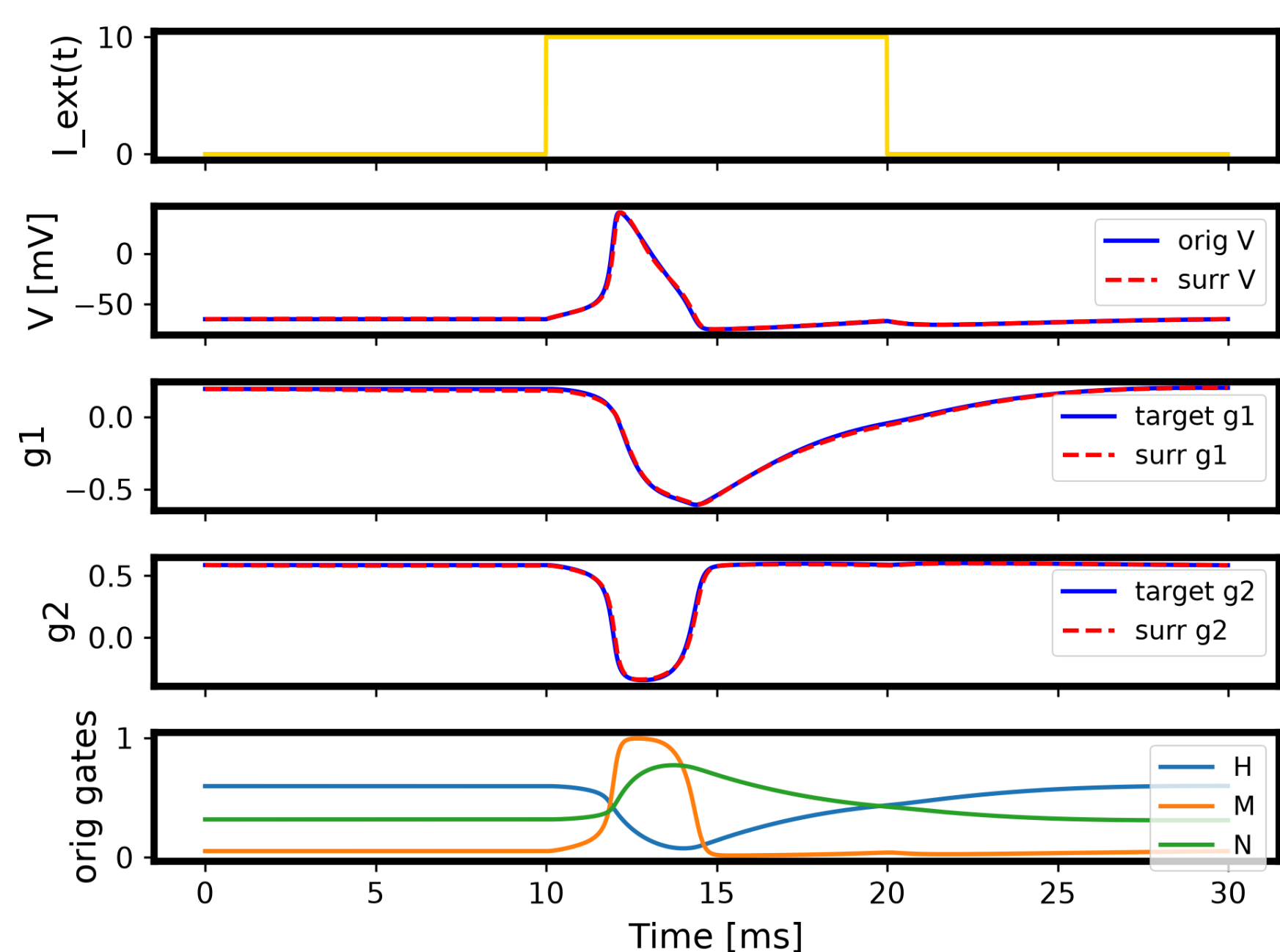
$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 1 & x & y & z & x^2 & xy & xz & y^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} z^5 \\ \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} \boldsymbol{\Xi}$$

...

Solve  $\dot{X} = \Theta(X)\boldsymbol{\Xi}$  (sparse regression) for  $\boldsymbol{\Xi}$  [4]

## Results and Discussion

### ① Proof of concept — single HH soma



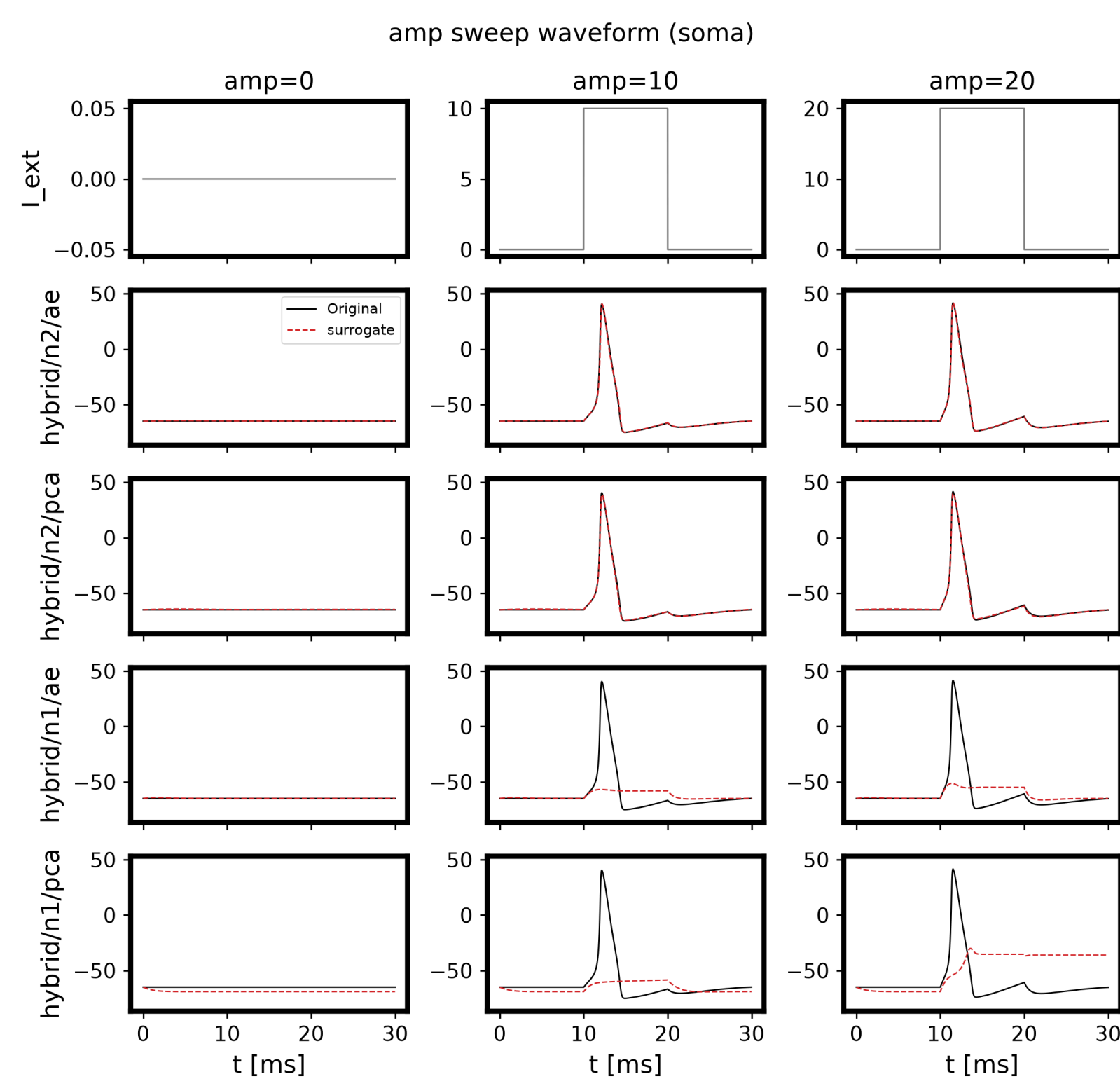
AE,  $n = 2$ , hybrid under a 10 ms pulse:  $V$ , the 2 AE latents  $g_1, g_2$ , and the 3 original gates. Original (blue) vs. surrogate (red, dashed) overlap.

### Waveform match

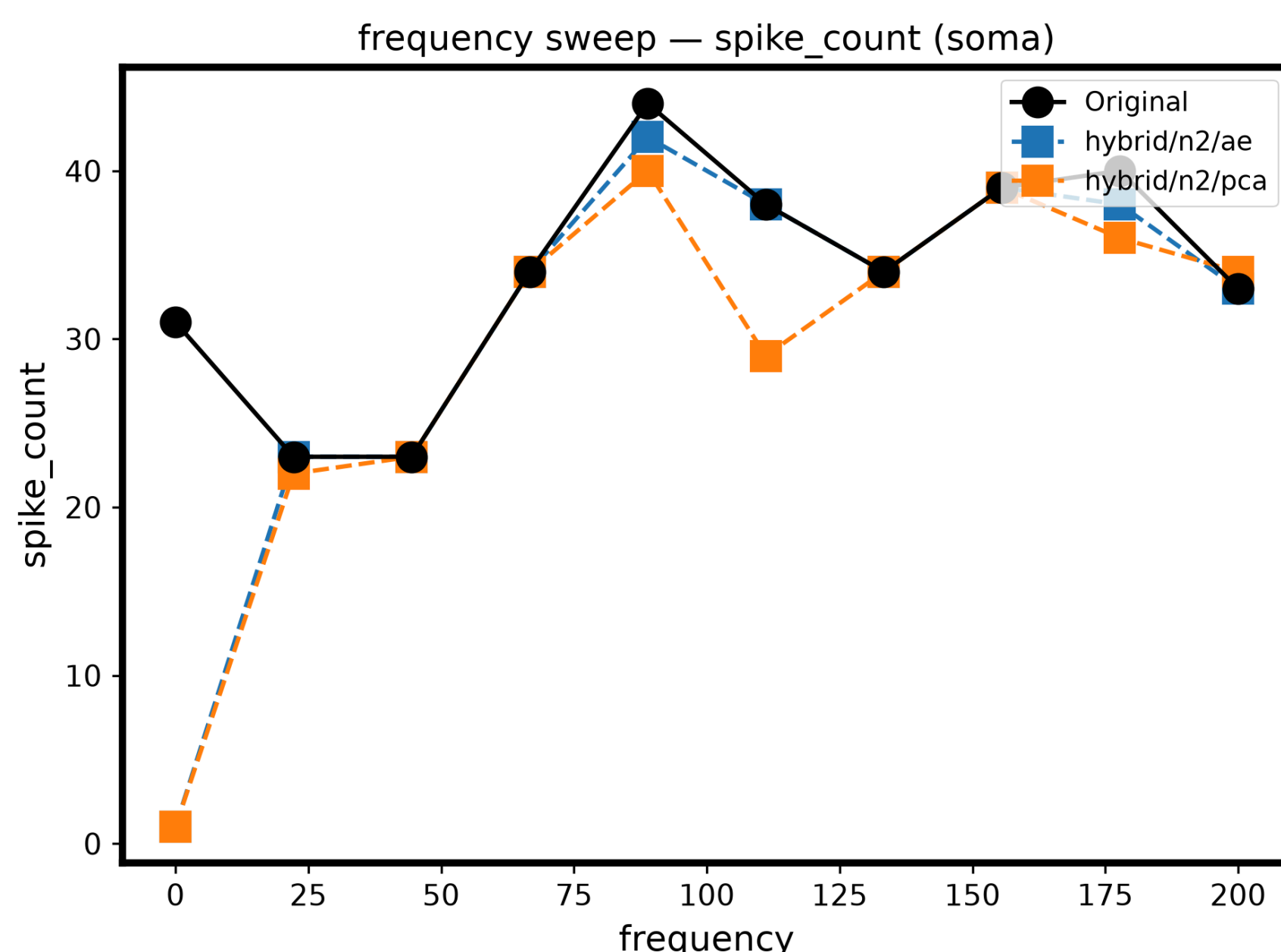
- RMSE **1.0 mV**, MAE **0.4 mV**, AP amp err **0.5 mV**
- spike count & latency **exact**

- 3 gates** → **2 latents**, spike reproduced with **near-zero error**.

### ② Generalization to unseen stimuli



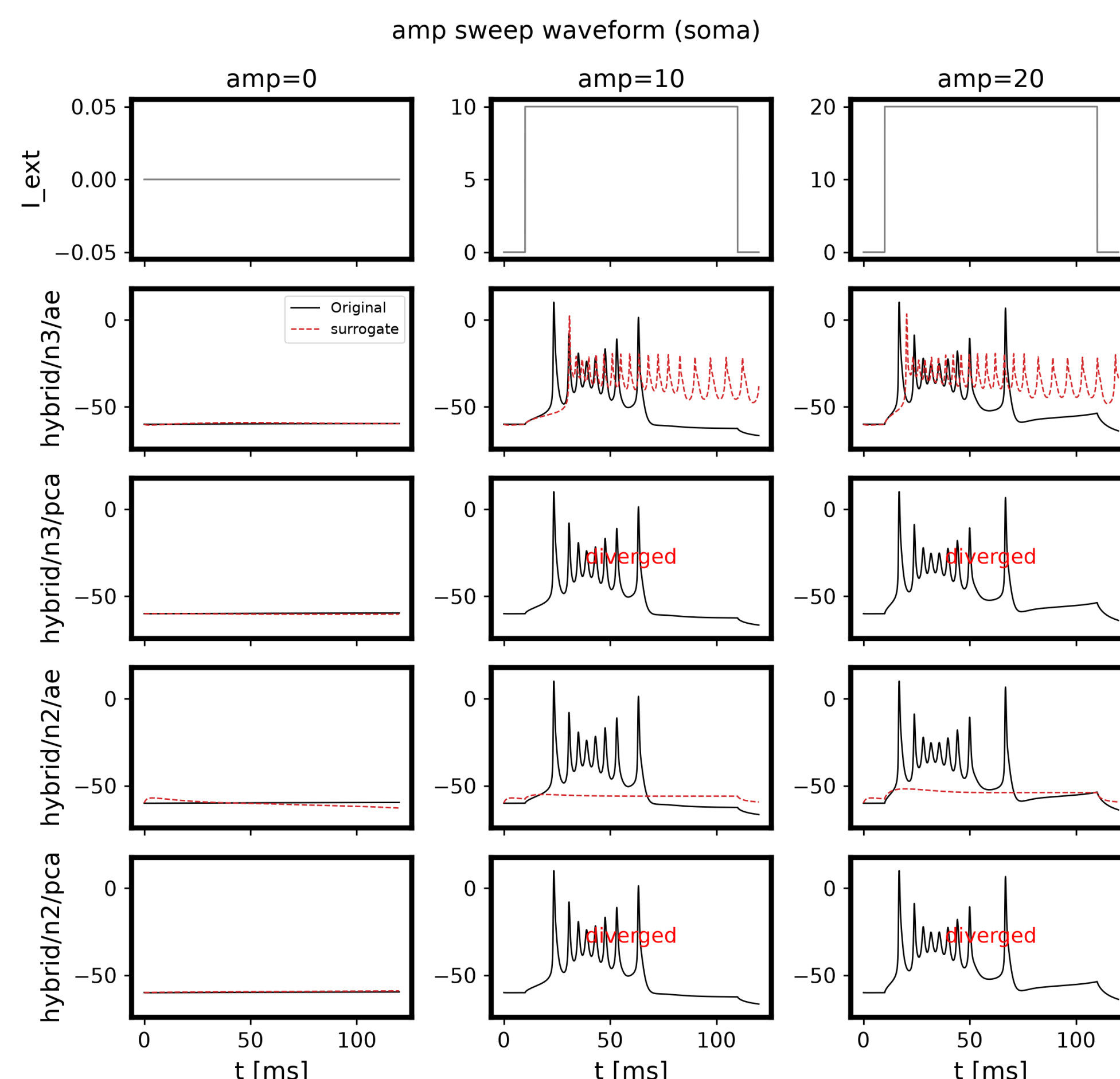
Amplitude sweep:  $n = 2$  spikes,  $n = 1$  collapses.



Frequency sweep under a **sinusoidal** drive unseen in training.

- Trained on pulses, **generalizes** to a sinusoidal drive (0–200 Hz);  $n = 2$  is the minimum latent dim.

### ③ Scaling up — Traub 19-compartment



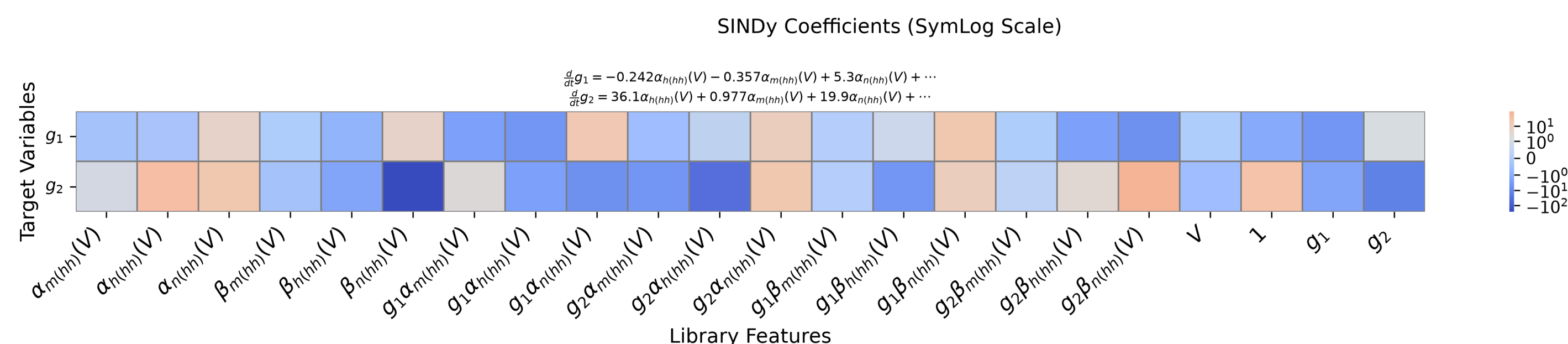
Spike count vs. amplitude, surrogate variants (soma replaced).

- Same framework transfers **without diverging**; **only AE +  $n = 3$  fires** (PCA,  $n = 2$  collapse).
- Gap:** spike count **overestimated** (9 vs. 6) — qualitative, not yet quantitative.

### Conclusion

- HH: near-zero error at **dim 4** → **2**, generalizes to sinusoid, **sparse latent eq**.
- Scales to **Traub 19-comp** without diverging; qualitative firing recovered.
- **Future:** close the spike-count gap, refine the physics-informed library.

- Code** — <https://github.com/MunechikaHaruki/SINDyNeuroSurrogate>
- [1] “錐体細胞.” Accessed: Dec. 08, 2025. [Online]. Available: <https://doi.org/10.14931/bsd.2091>
- [2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, “A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances,” *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.
- [3] 山崎 匡 and 五十嵐 潤, はじめての神経回路シミュレーション:1ニューロンからヒト全脳モデルまで. 森北出版株式会社, 2021, pp. 56–61.
- [4] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, “Data-Driven Discovery of Coordinates and Governing Equations,” *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.



**Sparse, interpretable identified model:** each  $\dot{g}_i$  is a few HH-rate-function terms.